

Miscellaneous/Review

* 1.

On this problem, only the answer will be graded.

- (a) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization $R_f = 0$?
- (b) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization R_f a field? (Recall that the zero ring is not a field.)
- (c) The Jordan form of an operator $A : \mathbb{C}^6 \rightarrow \mathbb{C}^6$ is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What is the Jordan form of the operator $A^3 - A$?

Solution 1.

- (a) $f = 6$ since it is the only element such that $f^t = 0$ in R for some power t
- (b) Recall that $R \rightarrow R_f$ is injective iff no power of f is a zerodivisor on R . In order for R_f to be a field, this map needs to not be injective. So, the zerodivisors on R (besides 6 since $R_6 = 0$) are our options. The zerodivisors are 2, 3, 4, 8, 9, 10. Also, R has two prime ideals (2) and (3). In order for R_f to be a field, it needs to have exactly one prime ideal, so either (2) or (3) needs to become the unit ideal in R_f . But if (2) is the only surviving prime ideal, then $(4) \subsetneq (2)$ is another ideal in the localization, which would mean that it is not a field. So, we need to localize at an element in (2) so that (3) is the only surviving prime ideal, and we need (3) to become the zero ideal in the localization. Choosing $f = 2, 4, 8, 10$ works since then $\frac{3}{1} = 0$ in the localization.
- (c)

$$A^3 - A = \begin{pmatrix} 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{which has Jordan form:} \quad \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

* 2.

On this problem, only the answer will be graded. Consider the symmetric group S_4 .

- (a) What is the order of S_4 ?
- (b) What is the order of its center $Z(S_4)$?
- (c) What is the order of its commutator subgroup $[S_4, S_4]$?
- (d) Is S_4 a simple group?
- (e) Is S_4 a solvable group?
- (f) How many conjugacy classes does S_4 have?

Solution 2.

- (a) 24
- (b) 1
- (c) 12 ($[S_4, S_4] = A_4$)
- (d) No, A_4 is a nontrivial normal subgroup.
- (e) Yes
- (f) 5

* 3.

On this problem, only the answers will be graded.

- (a) Give an example of a simple ring which is not a commutative ring.
- (b) Give an example a ring R , R -modules M_1, M_2, M_3 , and N and a short exact sequence $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ such that the corresponding sequence $0 \leftarrow \text{Hom}(M_1, N) \leftarrow \text{Hom}(M_2, N) \leftarrow \text{Hom}(M_3, N) \leftarrow 0$ is not exact.

Solution 3.

- (a) Let R be the ring of 2×2 matrices over $\mathbb{F}_2$. Then R is not commutative, and R is simple since the only nonzero two-sided ideal is R .
- (b) Let $R = \mathbb{Z}$ and consider $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$. Then applying $\text{Hom}(-, \mathbb{Z}/2\mathbb{Z})$ gives

$$0 \rightarrow \text{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} \rightarrow 0$$

which is not exact on the right since the map is multiplication by $2 = 0$, so it is not surjective.

* 4.

Let A be an algebra over $\mathbb{C}$ with two generators e and f and defining relations $e^2 = e, f^2 = f$.

- (a) Let B be the ring of 2×2 matrices over the polynomial ring $\mathbb{C}[t]$. Consider B as an (infinite-dimensional) algebra over $\mathbb{C}$. Show that there exists a unique $\mathbb{C}$ -algebra homomorphism $\phi : A \rightarrow B$ such that $\phi(e) = \begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix}$ and $\phi(f) = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.

- (b) Prove that the homomorphism ϕ from part (a) is injective.
- (c) Consider the algebra A as a vector space over $\mathbb{C}$. Write down a basis for A and prove that your answer is correct.

Solution 4.

- (a) We need to show that the map ϕ given is well-defined. So, we just need to show that it respects the relations in A , i.e. that $\phi(e^2 - e) = \phi(f^2 - f) = 0$. It is straightforward to compute that $\phi(e^2) = \phi(e)^2 = \phi(e)$ and similarly $\phi(f^2) = \phi(f)$, so this is a well-defined $\mathbb{C}$ -algebra homomorphism $A \rightarrow B$. It is unique because it is defined on the generators of A .
- (b) Any element in A can be written as a $\mathbb{C}$ -linear combination of $\{1, e, f, ef, fe, efe, fef, \dots\}$. So, let's think about the image of these elements under ϕ . We find that $\phi(efe) = t\phi(e)$, $\phi(fef) = t\phi(f)$, $\phi(efef) = t^2\phi(ef)$, $\phi(fe fe) = t^2\phi(fe)$, $\dots$, $\phi(efef \cdots e) = t^n\phi(e)$, $\phi(fe fe \cdots f) = t^n\phi(f)$, $\phi(efef \cdots f) = t^n\phi(ef)$, and $\phi(fe fe \cdots e) = t^n\phi(fe)$ for some n . Therefore, $\phi(a)$ for any $a \in A$ can be written as

$$\phi(a) = \begin{pmatrix} a_1 & 0 \\ 0 & a_1 \end{pmatrix} + p_e(t) \begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix} + p_f(t) \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} + p_{ef}(t) \begin{pmatrix} t & t \\ 0 & 0 \end{pmatrix} + p_{fe}(t) \begin{pmatrix} 0 & 0 \\ 1 & t \end{pmatrix}$$

where the $p_i(t) \in \mathbb{C}[t]$ are determined by the coefficients of the elements in a . Therefore, if $\phi(a) = 0$, we have

$$\begin{pmatrix} a_1 + p_e(t) + tp_{ef}(t) & t(p_e(t) + p_{fe}(t)) \\ p_f(t) + p_{fe}(t) & a_1 + p_f(t) + tp_{fe}(t) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solving this system gives that $p_e(t) = p_f(t) = -p_{ef}(t) = -p_{fe}(t)$, which then gives that $a_1 + (1 - t)p_e(t) = 0$. This is only possible if $p_e(t) = 0$, and then $a_1 = 0$. Thus, $a = 0$, and ϕ is injective.

- (c) A basis for A as a $\mathbb{C}$ -vector space is given by $\{1, e, f, ef, fe, efe, fef, \dots\}$, so this is an infinite-dimensional $\mathbb{C}$ -vector space.

*** 5.**

Let R be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(y, x)$.

- (a) Show that R is generated as a $\mathbb{C}$ -algebra by $x + y$ and xy .
- (b) Show that the map $\mathbb{C}[u, v] \rightarrow R$ sending u to $x + y$ and v to xy is an isomorphism.
- (c) Let S be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(-x, -y)$. Can S be generated as a $\mathbb{C}$ -algebra by two polynomials? Either give two generators, or prove that S is not generated by 2 polynomials.

Solution 5.

- (a) Let $P(x, y) \in R$. By induction on the power of P , we may assume that P is homogeneous of some degree n and that all lower power polynomials in R are generated by $x + y$ and xy over $\mathbb{C}$. Then we can write P as a constant times a product of linear factors, each of the form x, y , or $x + ay$ with $a \neq 0$. Because $P(x, y) = P(y, x)$, we must have that the number of x factors is the same as the number of y factors. So, if x^t divides P for some $t > 0$, then y^t divides P , but then $(xy)^t$ divides P , and we are done by the induction hypothesis. Therefore, we can assume that $t = 0$, i.e. no power of x or y divides P , and we can write $P(x, y) = c(x + a_1y) \cdots (x + a_ny)$ for some nonzero $c, a_i \in \mathbb{C}$. But then $P(y, x) = c(a_1x + y) \cdots (a_nx + y)$, and since $P(x, 1) = P(1, x)$, we have $(x + a_1) \cdots (x + a_n) = (a_1x + 1) \cdots (a_nx + 1)$. Comparing roots gives that $a_i = \frac{1}{a_i}$ for each i , so $a_i = \pm 1$ and so $P(x, y) = c(x + y)^r(x - y)^s$ for some $r + s = n$. Symmetry then implies that s is even, and we are done then since $(x - y)^2 = (x + y)^2 - 4xy$.
- (b) Part (a) gives that this map is surjective. So we just need to show injectivity. Suppose $Q(u, v) \in \mathbb{C}[u, v]$ is in the kernel, i.e. $Q(x + y, xy) = 0$, and assume that $Q(u, v) \neq 0$. Observe that taking $y = 0$ gives $Q(x, 0) = 0$, which implies $Q(u, 0) = 0$, so every term of Q has a factor of v . Similarly, taking $x = 0$ gives that $Q(y, 0) = 0$ and by symmetry in R , we have $Q(0, y) = 0$, so every term of Q has a factor of u as well. Therefore, by unique factorization, we have $Q(u, v) = uvQ'(u, v)$. Applying the map gives $(x+y)(xy)Q'(x+y, xy) = 0$, which implies that $Q'(x+y, xy) = 0$ since $(x+y)xy \neq 0$ and R is a domain. This means that $Q'(u, v)$ is an element of the kernel of strictly smaller degree than $Q(u, v)$. But then we could just repeat the argument with $Q'(u, v)$ to produce infinitely many elements of the kernel of smaller and smaller degree, which is impossible. Thus, $Q(u, v) = 0$.
- (c) We first show that $S = \mathbb{C}[x^2, xy, y^2]$. The backwards containment is clear. For the forward containment, note that if $P(x, y) \in S$, then $P(x, x) = P(-x, -x)$ is a polynomial in x where all terms must have even degree. Therefore, every term of $P(x, y)$ must have even total degree. So every monomial in $P(x, y)$ is of the form $x^i y^j$ with $i + j$ even and WLOG $i \geq j$. Then $x^i y^j = (xy)^j x^{i-j} \in \mathbb{C}[x^2, xy, y^2]$ since $i - j$ is also even. Then $S = \mathbb{C}[x^2, xy, y^2]$ cannot be generated as a $\mathbb{C}$ -algebra by two polynomials because it is 3-dimensional as an algebra over $\mathbb{C}$.

*** 6.**

Let R be a commutative ring and M an R -module. Recall that a prime ideal P is an associated prime of M if there exists some $m \in M$ such that P consists of those f such that $fm = 0$; i.e. P is the annihilator of some $m \in M$.

- (a) Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of R -modules, and let P be an associated prime of M . Prove that P is an associated prime of either M' or M'' .
- (b) Is the converse true? That is, if P is an associated prime of either M' or M'' , must it be the case that P is an associated prime of M ?

Solution 6.

- (a) Suppose P is not an associated prime of M' , and write $P = \text{ann}(m)$ for some $m \in M$. We want to show that $P = \text{ann}(m'')$ for some $m'' \in M''$. Let ϕ be the map $M \rightarrow M''$ in the SES. Then a natural choice is $m'' = \phi(m)$. We will show that this does indeed work. Let $f \in P$ so that $fm = 0$. Then $f\phi(m) = \phi(fm) = 0$, so $f \in \text{ann}(\phi(m))$. Thus, $P \subseteq \text{ann}(\phi(m))$. Conversely, if $f \in \text{ann}(\phi(m))$, then $\phi(fm) = f\phi(m) = 0$, so $fm \in \ker(\phi) = \text{im}(\varphi)$ by the exactness of the sequence where $\varphi : M' \rightarrow M$. Suppose towards a contradiction that $f \notin P$. Then $fm = \varphi(m') \neq 0$, so $m' \neq 0$ since φ is injective. We will show that this gives $P = \text{ann}(m')$, contradicting our assumption. For any $g \in P$, we have $\varphi(gm') = g\varphi(m') = g(fm) = f(gm) = 0$, so $gm' = 0$ by injectivity, which gives $g \in \text{ann}(m')$. On the other hand, given $h \in \text{ann}(m')$, we have $0 = \varphi(hm') = h\varphi(m') = hfm$, so $hf \in P = \text{ann}(m)$. Since $f \notin P$ and P is prime, we must have $h \in P$. This gives our contradiction; thus, $f \in P$ and $P = \text{ann}(\phi(m))$.
- (b) No, take for example the SES of $\mathbb{Z}$ -modules $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ where the first map is multiplication by 2. Then (2) is the annihilator of 1 in $\mathbb{Z}/2\mathbb{Z}$, but $\mathbb{Z}$ is torsion-free, so its only associated prime is (0).

* **7.**

Let L/K be a Galois extension of fields with Galois group G . Let S be the subset of roots of unity in L ; that is, S is the set of elements $x \in L$ such that $x^k = 1$ for some positive integer k .

- (a) Show that the action of G on L restricts to an action of G on S (i.e., the action of G preserves S).
- (b) Let $L = \mathbb{Q}(i)$ and $K = \mathbb{Q}$. Show in this case that G acts faithfully on S .
- (c) Give an example of L/K such that the action of G on S is not faithful.
- (d) Suppose that the action of G on S is transitive. Prove that $\text{char}(K) = 2$.

Solution 7.

- (a) The n th roots of unity in L are precisely the elements in L that are roots of the equations $t^n - 1$. But $t^n - 1 \in K[t]$, so G must permute the roots of this polynomial. Thus, G also acts on S .
- (b) In this case, $S = \{\pm 1, \pm i\}$ and since $[L : K] = 2$, $G \cong \mathbb{Z}/2\mathbb{Z}$. So the action of G on S is given by 0 acting trivially and 1 acting by $i \mapsto -i$. So this action is faithful since only the identity element in G acts trivially on S .
- (c) Let $L = \mathbb{Q}(\sqrt{2})$ and $K = \mathbb{Q}$. Then $S = \{\pm 1\}$ since those are the only roots of unity in $\mathbb{R}$ and $L \subseteq \mathbb{R}$. As in the previous example G is a cyclic group of order 2. But both elements of G fix S since $S \subseteq \mathbb{Q}$.
- (d) Imagine K has characteristic not equal to 2. Then -1 and 1 are distinct elements of $S \cap K$. But every element of G will fix both -1 and 1 since they are in K , so the action could not be transitive.